

Statistical Blendshape Calculation and Analysis for Graphics Applications

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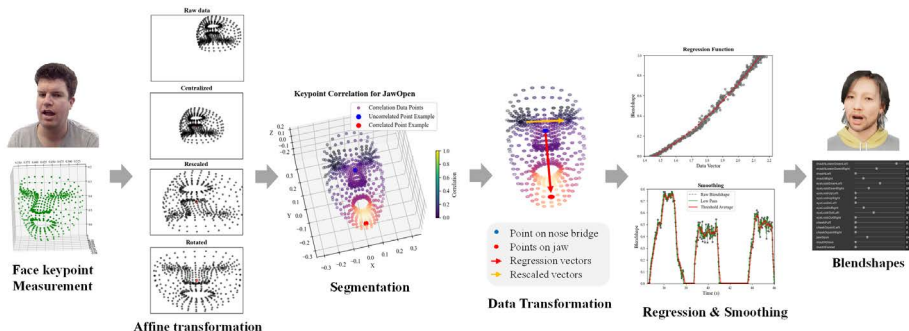


Figure 1: Logical process flow of blendshape estimation method

ABSTRACT

Real-time facial avatar animation is widely used in entertainment, office and other fields where blendshapes have become a common animation method. We independently developed an accurate blendshape prediction system for low-power VR applications using a webcam. Feature vectors are extracted through affine transformation and segmentation. Further transformation and regression analysis was used to develop statistical models with significant predictive power. Post-processing was used to further improve response stability, including smoothing filtering and nonlinear transformations. The system achieved accuracy similar to ARKit 6.

Index Terms: Blendshapes, character animation, statistical modeling, real time animation

1 INTRODUCTION

The rapid development of AI and VR fields, digital humans have a wide range of potential application. Facial capture and driving technology are some of the most important components needed for digital humans, which allow users to dynamically control the expression of virtual avatars in real time.

Human avatars are represented using a virtual mesh model. 3D mesh vertex positions need to be fully specified. Morphable mesh animation absorbs their advantages, limiting the computational cost while maintaining high accuracy. Among the many deformable methods, blendshapes are popular due to relative computational simplicity, physical interpretability and convenience. This work proposes a real-time blendshape animation method using a webcam shown in Fig. 1. Our contributions are:

- Independently developed statistical models using a webcam.
- Post-processing algorithms.
- Statistical testing showing similar performance with ARKit 6

2 METHODOLOGY

Our model converts face landmarks to blendshapes in real time using a standard webcam. To do this, the model performs:

$$f_b = F(R(T_d(S(T_a(X)))))) \quad (1)$$

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Where X represents the regressed facial landmarks and f_b are the blendshape estimates. The equation terms are:

- T_a : Affine Transformation (Fig. 1)
- S : Segmentation (Fig. 1)
- T_d : Data Transformation (Fig. 1)
- R : Regression (Fig. 2, Fig. 1)
- F : Smoothing (Fig. 2, Fig. 1)

To minimize regression error, we first performed a set of affine transformations. Affine transformations maintain collinearity between point sets, parallelism between lines, convexity between sets, vector magnitudes and geometric centroids. Transformations are chosen so landmarks are projected into a relatively rotation- and distortion-invariant basis.

After affine transformation, we use segmentation to reduce the dimensionality of the input data. We performed statistical analysis to prove that we were not losing correlated information while performing segmentation, maintaining high model accuracy.

Regression analysis was performed develop statistical models for each blendshape with significant predictive power. These models included different data transformation steps, including data and distribution changes needed to correct for limited heteroskedasticity. Examples included OLS, PCA, PLS, Gaussian, and ensemble methods. Statistical analysis and real-world testing demonstrated the need for additional nonlinear corrective terms were necessary to improve blendshape dynamic range, reduce bias effects between different faces and provide a better visual experience.

We also added smoothing/inter-frame correlation algorithms to reduce jitter and improve the smoothness of visualization for each blendshape. We used a filter approach to smoothing.

Two different data sources were used, 131 raw images taken with an iPhone 12 pro, and an unreal Engine 4 metahuman avatar that had been configured to receive LiveLink blendshape commands was used to create synthetic training data, usually 200-600 pictures at 400x600 pixels per blendshape. Four different training steps were required, feature section/segmentation, regression, weighting function, and smoothing.

3 RESULTS

For a more objective comparison we used a modified $F1$ scoring criteria based on test dataset annotations. These annotations

Table 1: Comparison Between ARKit and our methods, and response similarity between methods

Blendshape	ARKit			Our Model			Cross-Comparison					
	Pr	Re	F1[4]	Pr	Re	F1[4]	P Corr[2]	S Corr[3]	ξ [1]	Accuracy	MSD	Deviation
EyeBlinkLeft	1.00	1.00	1.00	1.00	1.00	1.00	0.576	0.572	0.305	0.752	0.061	0.216
EyeLookDownLeft	0.80	1.00	0.88	0.61	1.00	0.76	0.657	0.678	0.393	0.876	0.015	0.074
EyeLookInLeft	0.80	1.00	0.88	0.50	1.00	0.66	0.489	0.504	0.402	0.773	0.051	0.224
EyeBlinkRight	1.00	1.00	1.00	1.00	1.00	1.00	0.674	0.675	0.410	0.794	0.042	0.282
EyeLookDownRight	0.80	1.00	0.88	0.61	1.00	0.76	0.658	0.679	0.385	0.876	0.015	0.074
EyeLookInRight	0.83	1.00	0.90	0.80	0.80	0.80	0.567	0.606	0.400	0.914	0.007	0.133
JawOpen	1.00	1.00	1.00	1.00	1.00	1.00	0.921	0.886	0.682	0.965	0.001	0.039
MouthRight	0.88	1.00	0.94	0.80	1.00	0.88	0.636	0.537	0.349	0.961	0.001	0.025
MouthLeft	0.88	1.00	0.94	1.00	0.62	0.76	0.610	0.616	0.425	0.945	0.002	0.042
MouthSmileLeft	0.88	1.00	0.94	1.00	0.62	0.76	0.932	0.953	0.740	0.867	0.017	0.073
MouthSmileRight	0.88	1.00	0.94	0.80	1.00	0.88	0.907	0.943	0.702	0.861	0.019	0.067
MouthUpperUpLeft	0.88	1.00	0.94	1.00	0.62	0.76	0.474	0.467	0.407	0.925	0.005	0.078
MouthUpperUpRight	0.88	1.00	0.94	0.80	1.00	0.88	0.496	0.504	0.387	0.927	0.005	0.080
BrowDownLeft	0.77	1.00	0.87	0.63	1.00	0.77	0.699	0.544	0.404	0.698	0.091	0.337
BrowDownRight	0.77	1.00	0.87	0.75	0.85	0.80	0.717	0.518	0.373	0.756	0.059	0.235
Average	0.87	1.00	0.93	0.82	0.90	0.83	0.667	0.645	0.451	0.859	0.019	0.132

recorded when an expression could be observed to occur, but not specific expression intensity.

We display the correlation between our algorithm and ARKit as well as accuracy metrics of F1-score in Tab. 1. Overall F1-score, we reach to a small difference (9.53%) comparing to ARKit. Visual response comparison is shown in Fig. 4.

Testing was done using a 12th Gen Intel(R) Core(TM) i7-12700H CPU with 14 cores. 21 test videos are used for testing, with results shown in Fig. 3(a)-(c). Statistical analysis is shown with cumulative probability plots Fig. 3(d)-(f). The 50th and 95th percentile of CPU usage on 1 core range between 39.1% and 81.2%, total memory usage ranges between 549MB and 569MB. More than 90% of the utilized memory is consumed by the MediaPipe Holistic model. All processing steps are executed on the CPU. Algorithm response time 50th and 95th percentile values range between 0.86ms and 2.46ms.

4 CONCLUSION

We were able to successfully develop a statistical model to convert landmark estimates to blendshapes for 3D human avatar animation.

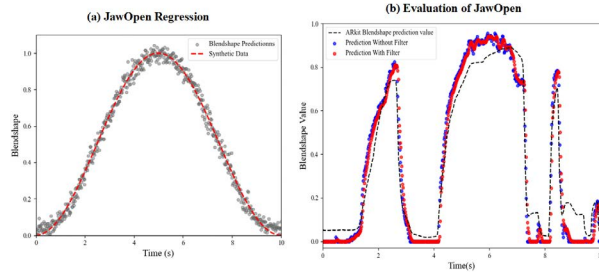


Figure 2: (a) JawOpen blendshape predictions (gray dots) compared with synthetic test data (dashed red). (b) Demonstrates the high degree of similarity between Apple ARKit 6 (dashed line) and the new model without filtering (blue) and with filtering (red).

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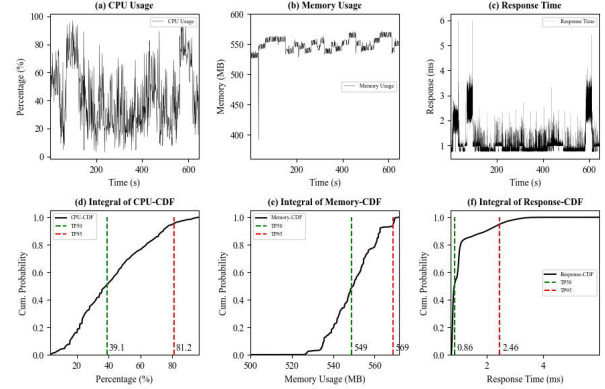


Figure 3: Performance validation data of 3D avatar algorithm. (a), (b), (c) The CPU usage/memory usage/response time over time. (d), (e), (f) CDF graph of (a), (b), (c), which shows the location of TP50/TP95 value.



Figure 4: Response comparison between raw image (left column), ARKit (middle column) and our algorithm (right column)

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